Data Warehousing Project Software Users Manual

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Revision Sheet

Revision	Date	Brief Summary of Changes
Version 1.0(draft)	January 16, 2026	Baseline document draft
Version 1.1	January 27, 2026	Expanded user documentation for data warehousing operations
Version 1.2	March 4th, 2026	Revised software commands section
Version 1.3	March 18th, 2026	Updated concept of operations section

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1. INTRODUCTION

This data warehousing project is a piece of a larger part of the "Algorithmic Trading and Short-term Forecast for Financial Time Series with Machine Learning Models" project. It is not designed with user experience in mind, as it is meant to be a middleman between input systems and output systems. User documentation may be limited in use due to this.

The purpose of this system is to load data retrieved by another system into a more accessible data warehouse, for use in a machine learning algorithm system, input. Access to the raw data may be more accessible to users via the previous system's data collection archives, however this documentation will describe how to access and operate the data warehouse system as a user.

1.1 Document Use

This Software User's Manual provides detailed operational guidance for users interacting with the Data Warehousing Project. The intended audience includes data warehouse engineers, system administrators, and technical users who require read or operational access to warehouse data.

The manual contains both instructional and reference material. Instructional sections describe common operational procedures, while reference sections provide background information, terminology, and troubleshooting guidance. Users are expected to have basic familiarity with relational database concepts and SQL querying.

1.2 Definitions and Acronyms

This section defines terminology and acronyms specific to the Data Warehousing Project to ensure consistent understanding across engineering, analytics, and operational teams.

1.2.1 Key Acronyms

- **DW** – Data Warehouse
- **DDL** – Data Definition Language
- **DML** – Data Manipulation Language
- **ETL / ELT** – Extract, Transform, Load / Extract, Load, Transform
- **SCD** – Slowly Changing Dimension
- **SLA** – Service Level Agreement

1.2.2 Key Definitions

- **Curated Data** – Data that has been cleaned, validated, and structured for analytical use.
- **Dimension Table** – A table containing descriptive attributes related to facts.
- **Fact Table** – A table containing quantitative data for analysis.
- **Incremental Load** – A loading strategy that processes only new or modified records.

- **Staging Area** – Temporary storage used during data transformation and validation.

1.3 References

[This section should provide a list of all references used in support of the development of the software users manual. Include a listing of all the documentation related to the product that is to be transitioned to the operations area, which includes any security or privacy protection consideration associated with its use. Also include as a part this, any licensing information for the product.]

2. Concept of Operations

The Data Warehousing Project functions as a centralized data repository that aggregates financial time-series data from upstream collection systems. Data is ingested on a scheduled basis, validated for quality, transformed into standardized schemas, and stored for downstream analytics and machine learning applications.

The system is designed to operate autonomously with minimal direct user interaction. Automated pipelines manage ingestion and transformation while users primarily consume curated datasets.

2.1 Operational Workflow

The system follows a structured pipeline to ensure data is ready for the machine learning models:

- **Data Ingestion:** Data is ingested from upstream sources on a strictly scheduled basis.
- **Quality Assurance:** Once ingested, data is validated for quality and transformed into standardized schemas.
- **Automated Management:** Automated pipelines manage the heavy lifting of ingestion and transformation.
- **Downstream Consumption:** The end goal is the storage of data for downstream analytics and machine learning applications.

2.2 User Interaction

Users primarily interact with the system as consumers of curated datasets. The system is not designed with a focus on "user experience" because its primary role is serving as a data intermediary. Consequently, user documentation focuses on technical access rather than a traditional graphical interface.

[This section should provide an overview of the software to include its intended use. Descriptions of any relevant business processes or workflow activities should be included. Any items required in support of the understanding of the software product should be included. This may require a description of theory, method, or algorithm critical to the effective use and understanding of the product.]

3. General Use

Users interact with the data warehouse using approved SQL clients or integrated analytics tools. Typical activities include validating data freshness, executing read-only analytical queries, reviewing record counts, and exporting result sets for reporting or research purposes.

Direct modification of warehouse tables by end users is not permitted to ensure data integrity and consistency across downstream systems.

[Information should be provided in support of routine user activities. It is important to identify actions that will be performed repetively to avoid redundancy within the users manual. For example, describing how to cancel, or interrupt, an operation while using the software would be in this section. Other task-oriented routine documentation could include: Software installation and deinstallation procedures, how to log on and off the application, and the identification of basic items/actions that are common across the applications' user interface.]

4. Procedures and Tutorials

This section provides step-by-step procedures for common operational tasks such as verifying scheduled data loads, performing validation queries, and exporting datasets. Users should ensure they have appropriate permissions prior to executing any procedures outlined in this manual.

[Information of a tutorial (ie., procedural) nature should be provided in the users documentation as clearly as possible. A consistent approach to the presentation of the material is important when trying to clearly communicate a concept to the user. The following structure is suggested:

Describe the purpose and concept for the tutorial information presented in the users manual. Include a list of all activities that must be completed prior to the initiation of the procedure or tutorial. Identify any material that should be used as reference in support of the task. List all cautions and other supporting information that are relevant in supporting the performance of the task.

It is important to list all instructional steps in the order that they should be performed, with any optional steps clearly identified. The steps should be consecutively numbered and the initial and last steps of the task should be clearly identified. It is important that the user understand how to successfully initiate and complete the procedure or tutorial.

Warnings and cautions should be distinguishable from instructional steps and should be preceded by a word and graphic symbol alerting the user to the item. For example, *warning (graphic),* would preceed a warning to the user. The use of the following format for warning and cautions is suggested: Word and graphic, brief description, instructional text, description of consequences, and proposed solution or workaround.]

5. Software Commands

The system supports ANSI-compliant SQL commands for data retrieval. Administrative or system-level commands are restricted to authorized personnel. Users are advised to avoid long-running or resource-intensive queries during scheduled data load windows. Doing so may cause the system to miss scheduled data insertion windows.

[The users manual should describe all software commands to include: required and optional parameters, defaults, precedence, and syntax. All reserved words and commands should be listed. This section should not only provide the commands, but should also provide examples of their use. Documentation should include a visual representation of the element, a description of its purpose, and an explanation of intended action. A quick reference card may be included in the users documentation provided the user with the ability to rapidly refer to commonly used commands.]

6. Navigational Features

Navigation within the data warehouse is based on schema organization and standardized naming conventions. Schemas correspond to logical processing layers such as raw, staging, and curated datasets, allowing users to quickly identify the appropriate data for their needs.

[The document should describe all methods of navigation related to the software application. All function keys, graphical user interface items, and commands used in support of application navigation should be described and supported with examples.]

7. Error Messages and Problem Resolution

Common issues include access permission errors, connection timeouts, incomplete data loads, and unexpected query results. Users should collect relevant error messages and query details when reporting issues to the data warehouse support team.

Known issues and resolutions are documented separately and updated as system changes occur.

[Information in support of problem resolution (ie., reference) should address all known problems or error codes present in the software application. Users should be provided information that will either help them recover from known problems, report unknown issues, and suggest application enhancements.]

Index

[An index provides an effective way for users to access documented information. It is important to remember that for an index to be useful it should contain words that users

are most likely to look up and should list all topics in the users documentation. Pay special attention to the granularity and presentation of the index topics. Place minor key words under major one, for example instead of using *files* with 30 pages listed, use *files, saving* and *files, deleting* with their associated specific pages listed.]